

# Training With Weightlifting Derivatives: The Effects of Force and Velocity Overload Stimuli

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## Abstract

Suchomel, TJ, McKeever, SM, and Comfort, P. Training with weightlifting derivatives: The effects of force and velocity overload stimuli. *J Strength Cond Res* 34(7): 1808–1818, 2020—The purposes of this study were to compare the training effects of weightlifting movements performed with (CATCH) or without (PULL) the catch phase of clean derivatives performed at the same relative loads or training without the catch phase using a force- and velocity-specific overload stimulus (OL) on isometric and dynamic performance tasks. Twenty-seven resistance-trained men completed 10 weeks of training as part of the CATCH, PULL, or OL group. The CATCH group trained using weightlifting catching derivatives, while the PULL and OL groups used biomechanically similar pulling derivatives. The CATCH and PULL groups were prescribed the same relative loads, while the OL group was prescribed force- and velocity-specific loading that was exercise and phase specific. Pre-intervention and postintervention isometric midthigh pull (IMTP), relative one repetition maximum power clean (1RM PC), 10-, 20-, and 30-m sprint, and 505 change of direction on the right (505R) and left (505L) leg were examined. Statistically significant differences in preintervention to postintervention percent change were present for relative IMTP peak force, 10-, 20-, and 30-m sprints, and 505L (all  $p < 0.03$ ), but not for relative 1RM PC or 505R ( $p > 0.05$ ). The OL group produced the greatest improvements in each of the examined characteristics compared with the CATCH and PULL groups with generally moderate to large practical effects being present. Using a force- and velocity-specific overload stimulus with weightlifting pulling derivatives may produce superior adaptations in relative strength, sprint speed, and change of direction compared with submaximally loaded weightlifting catching and pulling derivatives.

**Key Words:** isometric midthigh pull, power clean, strength, sprinting, change of direction

## Introduction

Researchers have demonstrated that weightlifting movements may provide a superior strength-power training stimulus compared with jump training (50,51), traditional resistance training (25), and kettlebell training (31). One reason for these training effects may be due to the similarity between the second pull of weightlifting movements and the coordinated triple extension of the hips, knees, and ankles (plantar flexion) that occurs during the propulsive phases of jumping, sprinting, and change of direction tasks (26). In addition, weightlifting movements may provide a superior overload stimulus compared with other training methods given their requirement to move moderate to heavy loads with ballistic intent. In fact, researchers have indicated that weightlifting movements and their derivatives produce greater power outputs compared with the majority of other resistance training exercises (38). Thus, given their potential to improve strength-power performance, it is not surprising that many practitioners implement the weightlifting movements and their derivatives within resistance training programs (22,33).

Weightlifting movements and their derivatives are traditionally implemented by practitioners to include the catch phase of the movement. Although weightlifting catching derivatives (i.e., those that remove an aspect of the full

weightlifting movement) have been shown to produce positive strength-power training effects and load absorption benefits, more recent literature has indicated that weightlifting pulling derivatives (i.e., those that exclude the catch phase) may provide a comparable (4,5) or superior (27,28,47–49) training stimulus compared with weightlifting catching derivatives with regard to peak force, velocity, power, rate of force development, impulse, and work. Despite the existence of several cross-sectional studies, only one study has compared the effects of longitudinal training with either weightlifting catching or pulling derivatives. Comfort et al. (7) indicated that there was no statistical or practically meaningful difference between training 2 times per week for 8 weeks with either weightlifting catching or pulling derivatives on rapid force development during isometric midthigh pulls (IMTPs), one repetition maximum power clean (1RM PC) performance, squat jumps, or countermovement jumps. Although these findings are important, it should be noted that the loading between the catching and pulling derivative groups was identical (volume and relative loads matched), which may partly explain the similarity in the observed adaptations. Thus, further research is needed to determine whether differences in loading produce unique performance adaptations.

Weightlifting pulling derivatives may provide a greater force and velocity overload stimulus compared with catching derivatives (39,40). Although practitioners are limited to prescribing up to the 1RM of weightlifting catching derivatives, pulling derivatives may benefit force production (i.e., strength)

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characteristics to a greater extent due to their ability to use loads in excess of an athlete's 1RM PC. For example, some pulling derivatives, such as the midhigh pull and countermovement shrug may be loaded up to 140% of 1RM PC (8,9,30). In addition to greater potential force production, pulling derivatives, such as the jump shrug and hang high pull, produce greater movement velocities (49), which may benefit rapid force production characteristics. Based on the kinetic similarities between weightlifting catching and pulling derivatives presented in a recent study (7), it is possible that both modes of training (inclusion or exclusion of the catch phase) may be implemented to enhance an athlete's performance. However, it is also possible that superior training benefits may be displayed if a force- (e.g., loads in excess of catching derivative 1RM) and velocity-specific (e.g., greater velocities through more ballistic exercises) overload stimulus is provided with weightlifting pulling derivatives. Thus, further research is needed to explore this notion to better inform strength training prescription. The purposes of this study were to compare the training effects of weightlifting movements performed with (CATCH) or without (PULL) the catch phase of clean derivatives performed at the same relative loads or training without the catch phase using a force- and velocity-specific overload stimulus (OL) on isometric and dynamic performance tasks. In line with previous research (7), it was hypothesized that there would be no statistical or practically meaningful differences between the CATCH and PULL groups. However, it was also hypothesized that the OL group would demonstrate the greatest adaptations in isometric and dynamic performance compared with both the CATCH and PULL groups.

## Methods

### Experimental Approach to the Problem

To examine the differences in isometric and dynamic performance enhancement after resistance training programs that used weightlifting catching or pulling derivatives, a repeated-measures between-group design was used. The subjects completed 10 weeks of training (3 times per week) and were assessed before the training intervention and again after 10 weeks of training (Figure 1). Changes in isometric and

dynamic performance were assessed using the IMTP and a 1RM PC, 30-m sprints, and 505 change of direction.

### Subjects

Male collegiate athletes (ranged in age from 18–28 years old) and resistance-trained men with previous experience with the PC and its derivatives were recruited to participate in this study. Twenty-nine subjects volunteered and were randomly assigned to either the CATCH, PULL, or OL group. Two subjects voluntarily withdrew from the study, one because of an injury sustained during intramural sports outside of the study, and the other due to a desire to train more than 3 days per week. The characteristics of the subjects in each group are displayed in Table 1. All subjects who completed the study attended 100% of the training sessions. Before their participation, each subject read and signed a written informed consent form, in accordance with the Carroll University's institutional review board.

An a priori power analysis was completed using G\*Power (version 3.1.9.2). At a power level of 0.90, for an a priori alpha level of  $\leq 0.05$ , it was determined that at least 24 subjects were needed to display at least moderate effect sizes (Hedge's  $g \geq 0.50$ ) between groups, based on previous findings (12).

### Procedures

As displayed in Figure 1, preintervention and postintervention testing was completed over the course of 2 testing sessions separated by 48–72 hours to decrease the overall volume of tests as well as to accommodate the subjects' schedules. The time between the 2 postintervention sessions was kept consistent with the 2 preintervention testing sessions. In addition, a minimum of 48 hours of recovery was required before the subjects' testing sessions. Each testing session was scheduled to take place within 2 hours of subjects' preintervention testing sessions to account for changes in circadian rhythm. Before each testing session, the subjects performed the same standardized warm-up that consisted of stationary cycling, dynamic stretching, body weight squats, and progressive vertical jumps (45,47,48).

**Isometric Midhigh Pull Assessment.** The methodology used for IMTP testing has been previously described (3). Briefly, each

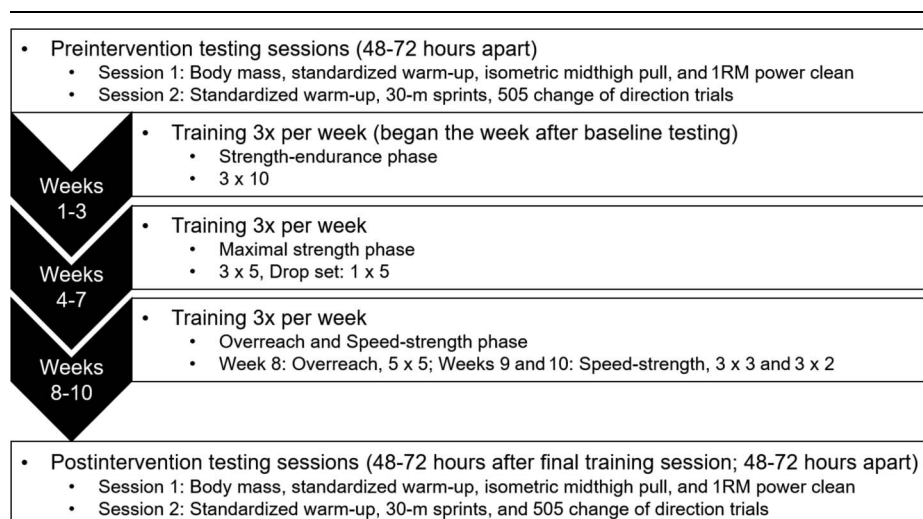


Figure 1. Testing and training sequence.

**Table 1**  
**Subject demographics for each training group.\*†**

|                                                 | CATCH (n = 9) | PULL (n = 9) | OL (n = 9)  |
|-------------------------------------------------|---------------|--------------|-------------|
| Age (y)                                         | 22.8 ± 3.6    | 22.2 ± 2.3   | 22.3 ± 1.2  |
| Body mass (kg)                                  | 85.8 ± 13.4   | 84.3 ± 17.3  | 83.0 ± 13.6 |
| Height (cm)                                     | 180.8 ± 5.8   | 179.6 ± 3.7  | 173.4 ± 9.3 |
| Power clean experience (y)                      | 7.2 ± 3.7     | 6.4 ± 2.4    | 6.4 ± 1.8   |
| Relative 1RM power clean (kg·kg <sup>-1</sup> ) | 1.20 ± 0.16   | 1.19 ± 0.18  | 1.25 ± 0.15 |
| Relative 1RM squat (kg·kg <sup>-1</sup> )       | 1.75 ± 0.40   | 1.73 ± 0.17  | 1.76 ± 0.32 |

\*CATCH = catch group; PULL = pull group; OL = overload group; 1RM = one repetition maximum.

†Relative 1RM squat strength was estimated using the subjects' heaviest loads lifted, sets, and repetitions of their most recent training phase completed before the start of the training program.

subject was positioned within an adjustable IMTP rig (Kairos Strength, Murphy, NC). An immovable barbell (Werksan Olympic Bar, Werksan, Moorsetown, NJ) was positioned at a height, which replicates the start of the second pull phase of the clean, resulting in knee and hip angles between 125–135° and 140–150°, respectively, based on previous recommendations (6). Individual angles were recorded and replicated during the postintervention testing session. In accordance with previous methods (3), the subjects' hands were strapped and taped to the barbell to prevent grip from being a limiting factor. After being given instructions regarding the countdown procedures, each subject performed 2 submaximal pulls, with one each at 50 and 75% of their perceived maximal effort, separated by one minute of rest. After a 2-minute rest period, each subject performed the first of at least 2 maximal effort pulls.

Before the maximal effort pulls, subjects were given final instructions. Specifically, the subjects were instructed to pull “as fast and hard as possible” and “push their feet down into the force plates.” After being instructed to, subjects first positioned their feet on the dual force plates (PASPORT force plate; PASCO Scientific, Roseville, CA) located under the immovable barbell. Next, the subject was instructed to get into their “ready position,” which was the previously measured starting position. The subjects were then instructed to remove any slack in their arms with the cue “tension on the bar.” Once the subjects' body position was stabilized (verified by watching the force trace), the subject was given a countdown of “3, 2, 1, Pull!” Each IMTP trial was performed for approximately 5 seconds, and strong verbal encouragement was provided. Subjects performed 2 maximal IMTP trials with 2 minutes of rest between trials. If the difference in peak force between the trials was greater than 250 N, or a visible countermovement was performed before the pull, a third trial was performed (3,6). The vertical ground reaction force data for the IMTP trials was recorded by the force plates sampling at 1,000 Hz.

**One Repetition Maximum Power Clean.** The 1RM PC of each subject was established using previously discussed methods (49). A self-selected warm-up with a 20-kg barbell was followed with warm-up PC sets using submaximal loads (e.g., 5 repetitions at 30 and 50%, 3 repetitions at 70%, and one repetition at 90% 1RM). During the preintervention testing session, subjects warmed up using percentages of their estimated 1RM PC, while percentages of the 1RM established during the preintervention session were used within the warm-up during the postintervention session. After the final warm-up repetition, the principal investigator and the subject

determined each maximal attempt load. A minimum 2.5 kg increase was required, and loads were progressively increased until a failed attempt occurred. Subjects were given at least 3 minutes of rest in between 1RM attempts. Any PC repetition caught with the top of the subject's thigh below parallel was ruled as an unsuccessful attempt. This was visually monitored during each 1RM attempt.

**Sprint Performance.** Thirty-meter sprint performance, with splits at 10- and 20-m, was assessed on an indoor track surface in the university's athletic fieldhouse using laser timing gates, which were positioned at approximately hip height (Brower Timing Systems, Draper, UT). After the standardized warm-up, each subject completed submaximal warm-up sprints at 50, 75, and 90% of their perceived maximum effort. The subjects were positioned 30 cm behind a marked starting line to prevent an inadvertent triggering of the timing system. After the last warm-up sprint, subjects received a 2- to 3-minute rest period before completing maximum effort sprints. Each subject performed two 30-m sprints with 3 minutes of rest between each sprint. However, a third sprint was performed if a tenth of a second difference existed between each sprint. All sprints were performed using a staggered, two-point static starting stance. The principal investigator demonstrated the starting position, and the subjects were asked to refrain from any preparatory movement (e.g., rearward sway) before the start of each sprint.

**Change of Direction Performance.** After a self-selected rest period after the 30-m sprints, subjects completed the 505 test to assess change of direction performance (1). Timing gates (Brower Timing Systems) and cones were set up 10- and 15-m from the start line, respectively. Subjects lined up in a staggered stance and ran 15 m crossing through the timing gates at 10 m, made a 180° turn at 15 m, and ran 5 m back through the timing gates. Foot placement during the 180° turn was visually monitored during each trial. Before the maximal trials, each subject performed completed a warm-up at 75% of their perceived maximum. The subjects then performed 3 maximal effort repetitions each, cutting with both their right (505R) and left (505L) legs, with one minute of rest between trials. The order of which leg was used for cutting was randomized during the preintervention testing session and kept consistent for each individual subject throughout the study.

**Training Intervention.** As mentioned above, each group trained 3 days per week for 10 weeks under the supervision of a certified strength and conditioning coach. The program was modified

from a recent review article that provided 18 weeks of programming with weightlifting derivatives in accordance with each group (39). Each weightlifting catching and pulling derivative was programmed based on the 1RM PC achieved during the preintervention testing session, similar to previous research (7,9,35,37,49). In addition, all weightlifting derivatives prescribed within the training program were coached using the technique described within previous literature (17–19,36,41,42). Nonweightlifting derivative exercises were added to the training intervention to increase the ecological validity of each program because weightlifting movements are rarely programmed in isolation for nonweightlifting athletes. Before the start of the training program, each subject provided the heaviest loads lifted, sets, and repetitions for the nonweightlifting derivative exercises (e.g., back squat, bench press, bent-over row, etc.) during their most recent training sessions. The 1RM for each exercise was then estimated, and the relative loads (Table 2) were determined using the set-repetition best method as discussed within previous literature (15,16). Using this method of loading, relative loads were based on percentages of the RM of the prescribed repetitions. For example, 90% of 3 sets of 10 repetitions use 90% of the subject's estimated 10RM weight. However, although a range of loads was prescribed, this method of loading also allowed the subjects to gauge the appropriate loads based on how many repetitions they feel that they could have performed beyond the prescribed number of repetitions (16). It should be noted that the 1RM for each nonweightlifting derivative exercise was recalculated throughout the study based on the loads that were performed in training. Finally, weightlifting derivatives prescribed using 3 sets of 10 repetitions were programmed using cluster sets of 5 repetitions with 30–40 seconds of intraset rest based on previous recommendations (23).

The differences between the training programs were that the CATCH group trained using PC derivatives with the catch phase during every repetition, while the PULL and OL groups trained using biomechanically similar PC derivatives that removed the catch phase (Table 3). The PULL group performed their derivatives with the same relative load as the CATCH group based on their 1RM PC (e.g., CATCH = PC at 80% 1RM; PULL = clean pull from the floor at 80% 1RM). This was performed to match the volume-load between the CATCH and PULL groups. By contrast, the OL group performed their PC derivatives with either a force or velocity overload stimulus, using either heavier (e.g., CATCH = PC at 80% 1RM; OL = clean pull from the floor with 100% 1RM) or lighter loads (e.g., CATCH = hang PC at 65% 1RM; OL = jump shrug at 30% 1RM), respectively. The velocity overload stimulus was also provided by prescribing pulling derivatives that are more ballistic in nature (e.g., jump shrug) (47–49). Although the volume-load was different between the OL group and other groups, this was performed to increase the ecological validity of prescribing pulling derivatives in line with previous recommendations (39). Further detail on

**Table 2**

**Relative loading based on set-repetition best for the nonweightlifting derivative exercises.\***

| Week: | Volume (sets × repetitions) | Day 1 | Day 2 | Day 3 |
|-------|-----------------------------|-------|-------|-------|
| 1     | 3 × 10                      | 80%   | 80%   | 70%   |
| 2     | 3 × 10                      | 85%   | 85%   | 75%   |
| 3     | 3 × 10                      | 90%   | 90%   | 80%   |
| 4     | 3 × 5                       | 85%   | 85%   | 70%   |
| 5     | 3 × 5                       | 90%   | 90%   | 75%   |
| 6     | 3 × 5                       | 95%   | 95%   | 77.5% |
| 7     | 3 × 5                       | 80%   | 80%   | 65%   |
| 8     | 5 × 5                       | 85%   | 85%   | 75%   |
| 9     | 3 × 3                       | 90%   | 90%   | 77.5% |
| 10    | 3 × 2                       | 85%   | 85%   | 75%   |

\*Relative loads were based on percentages of the repetition maximum (RM) of the prescribed repetitions (e.g., 90% of 3 × 10 uses 90% of the subject's estimated 10RM weight). Relative intensities were described as very light (65–70%), light (70–75%), moderately light (75–80%), moderate (80–85%), moderately heavy (85–90%), heavy (90–95%), and very heavy (100%) (15).

the relative load progression for the weightlifting derivatives of each training group is displayed in Table 4.

**Data Analyses.** A laptop computer and specialist software (PASCO Capstone, PASCO Scientific) were used to directly record force-time data during the IMTP trials. Because low-pass filtering procedures may underestimate IMTP kinetics (20), unfiltered data were used for data analysis. The force-time data of each trial were exported to and graphed in Microsoft Excel (Microsoft Corp., Redmond, WA). Each subject's body mass in Newtons was subtracted from the force-time data, to provide net force, and the maximum force recorded from the force-time curve during the IMTP trials was recorded as the peak force. The average of the 2 most similar trials, with regard to peak force production, was used for statistical comparisons. Finally, relative peak force was calculated by dividing the peak force of each IMTP trial by each subject's body mass that was recorded during each testing session. Similar to IMTP peak force, relative 1RM PC data were determined by dividing the 1RM PC of each subject by their body mass during each respective testing session. For sprinting performance, 10-, 20-, and 30-m times were recorded during each sprint. The average time of the 2 sprints was used for statistical analysis. In the event that the subject had to complete a third sprint, the average of the 2 most similar times was used for comparison. Similar to the sprints, the average of the 2 most consistent times for the 505R and 505L change of direction performances was used for statistical analysis. The percent change of each subject was calculated from pre-intervention to postintervention by using the below equation. The average of the individual percentage changes was then used to assess the changes of each group throughout the study.

$$\text{Percent change in performance (\%)} = \left( \frac{[\text{New score} - \text{Old score}]}{[\text{Old score}]} \right) \times 100$$



**Table 3****Ten-week resistance training program.\*†**

| Training block           | Day 1                                                      | Day 2                                                                                                                    | Day 3                                                                                               |
|--------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Strength-endurance       | Back squat<br>Military press<br>Split squat<br>Bench press | Power clean from floor/clean pull from floor‡<br>Stiff-legged deadlift<br>Bent-over row<br>Pull-up                       | Power clean from floor/clean pull from floor‡<br>Back squat<br>Incline bench press<br>Bent-over row |
| Max-strength + overreach | Push press<br>Back squat<br>Bench press<br>Lunge           | Midthigh power clean/midthigh pull‡<br>Power clean from floor/clean pull from floor‡<br>Stiff-legged deadlift<br>Pull-up | Midthigh power clean/midthigh pull‡<br>Back squat<br>Incline bench press<br>Dumbbell row            |
| Speed-strength           | Jerk<br>1/4 squat +<br>Squat jump<br>Bench press           | CM power clean/<br>CM shrug‡<br>Hang power clean/<br>Hang high pull‡                                                     | Jerk<br>Hang power clean/<br>Jump shrug‡                                                            |

\*CM = countermovement.

†1/4 squats were performed using a concentric-only movement off of the safety bars of a squat rack from a knee angle of 115–125°, and squat jumps were performed from a knee angle of approximately 90°.

‡Weightlifting pulling derivative prescribed for the pull and overload groups.

Finally, the weekly volume-load and pre-post intervention volume-load completed by each group was calculated as the product of sets, repetitions, and load.

**Statistical Analyses**

Normality of all data was examined using the Shapiro-Wilk test of normality. The criteria for the removal of outliers were if a data

point was greater than 3 times the *SD* of that specific test. However, because the sprinting data all took place as part of the same test, outliers were removed from all sprint test comparisons. Levene's test was used to assess the heterogeneity of variance between groups. Test-retest reliability was assessed during each testing session using two-way mixed intraclass correlation coefficients (ICCs) and typical error expressed as a coefficient of variation percentage (CV%). The ICCs were interpreted as poor

**Table 4****Clean derivative relative load progression for each training group.\***

| Training group | Training block           | Clean derivative(s)                               | Load progression (% 1RM power clean)                                                                                                        |
|----------------|--------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| CATCH          | Strength-endurance       | Power clean from floor                            | Wk 1: 55–57.5%<br>Wk 2: 57.5–60%<br>Wk 3: 60–62.5%                                                                                          |
|                | Max-strength + overreach | Midthigh power clean, power clean from floor      | Wk 4: 55–60%, 70–75%<br>Wk 5: 60–65%, 75–80%<br>Wk 6: 65–70%, 80–82.5%<br>Wk 7: 50–55%, 65–70%<br>Wk 8: 53–58%, 63–68%                      |
|                | Speed-strength           | Countermovement power clean, hang power clean     | Wk 9: 60–65%, 70–75% (Day 2) and 55–60% (Day 3)<br>Wk 10: 55–60%, 65–70% (Day 2) and 50–55% (Day 3)                                         |
|                |                          |                                                   |                                                                                                                                             |
| PULL           | Strength-endurance       | Clean pull from floor                             | Wk 1: 55–57.5%<br>Wk 2: 57.5–60%<br>Wk 3: 60–62.5%                                                                                          |
|                | Max-strength + overreach | Midthigh pull, clean pull from floor              | Wk 4: 55–60%, 70–75%<br>Wk 5: 60–65%, 75–80%<br>Wk 6: 65–70%, 80–82.5%<br>Wk 7: 50–55%, 65–70%<br>Wk 8: 53–58%, 63–68%                      |
|                | Speed-strength           | Countermovement shrug, hang high pull, jump shrug | Wk 9: 60–65%,<br>70–75%, 55–60%<br>Wk 10: 55–60%,<br>65–70%, 50–55%                                                                         |
|                |                          |                                                   |                                                                                                                                             |
| OL             | Strength-endurance       | Clean pull from floor                             | Wk 1: 75–77.5%<br>Wk 2: 77.5–80%<br>Wk 3: 80–82.5%                                                                                          |
|                | Max-strength + overreach | Midthigh pull, clean pull from floor              | Wk 4: 110–120%, 90–95%<br>Wk 5: 120–127.5%, 95–100%<br>Wk 6: 127.5–135%, 100–102.5%<br>Wk 7: 112.5–120%, 85–87.5%<br>Wk 8: 107–112%, 80–85% |
|                | Speed-strength           | Countermovement shrug, hang high pull, jump shrug | Wk 9: 105–110%, 40–45%, 35–40%<br>Wk 10: 100–105%, 35–40%, 30–35%                                                                           |
|                |                          |                                                   |                                                                                                                                             |

\*CATCH = catch group; PULL = pull group; OL = overload group.

(<0.50), moderate (0.50–0.74), good (0.75–0.90), and excellent (>0.90) (29). Acceptable within-session variability was classified as <10% (11). A series of one-way analysis of variance with Bonferroni *post hoc* analyses were used to examine the percent change differences in preintervention to postintervention relative IMTP peak force, relative 1RM PC, 10-, 20-, and 30-m sprint time, 505 change of direction times, and volume-load between the CATCH, PULL, and OL groups. A criterion *p*-value of ≤0.05 was used to identify statistical significance. In addition, the magnitude of any changes was determined through the calculation of effect sizes (Hedge's *g*). Effect sizes were interpreted based on the “highly trained” status (i.e., individuals training for at least 5 years) outlined in previous literature (32). Specifically, effect sizes were interpreted as trivial, small, moderate, and large when magnitudes were <0.25, 0.25–0.49, 0.50–1.0, and >1.0, respectively. All statistical analyses were performed using SPSS (Version 25, IBM, New York, NY).

## Results

All percent change data were normally distributed and demonstrated similar variance within each group. The reliability of all testing data from each testing session ranged from good to excellent (ICC = 0.75–0.99) with acceptable variability (CV% = 0.5–3.6%) for each group. The descriptive testing data and volume-load data of each group are displayed in Tables 5 and 6, respectively. Statistically significant differences in preintervention to postintervention percent change were present for relative IMTP peak force (*p* = 0.005), 10- (*p* = 0.023), 20- (*p* = 0.028), and 30-m sprints (*p* = 0.028), and 505L (*p* = 0.018), but not for relative 1RM PC (*p* = 0.369) or 505R (*p* = 0.405). Furthermore, no statistically significant differences existed between groups for weekly (*p* = 0.288–0.998) or total volume-load (*p* = 0.331). Individual data and effect size comparisons between groups are displayed in Figures 2–5.

*Post hoc* analysis revealed that the OL group produced statistically greater relative IMTP peak force improvements compared with the CATCH group (*p* = 0.005, *g* = 1.64), but not the PULL group (*p* = 0.931, *g* = 0.43). There was also no statistical difference between the CATCH and PULL group (*p* = 0.056, *g* = 1.21). Regarding sprint performance, *post hoc* analysis revealed that 10-m improvements were greater for the OL group compared with the CATCH group (*p* = 0.026, *g* = 1.32), but not the PULL group (*p* = 0.121, *g* = 1.35). Furthermore, no statistical difference in 10-m sprint improvements existed between the CATCH and PULL group (*p* = 1.000, *g* = 0.29). Although the OL group produced the greatest improvements in 20- and 30-m sprint performance, these differences were not statistically different from the CATCH (*p* = 0.056, *g* = 1.17; *p* = 0.065, *g* = 1.10) or PULL groups (*p* = 0.064, *g* = 1.26; *p* = 0.053, *g* = 1.44). No statistical difference existed between the CATCH and PULL groups for either variable (both *p* = 1.000, *g* = 0.03–0.04). Finally, *post hoc* analysis for the 505L test revealed that the OL group produced greater improvements compared with the CATCH group (*p* = 0.017, *g* = 1.29), but not the PULL group (*p* = 0.178, *g* = 0.69). No statistical differences were present between the CATCH and PULL groups (*p* = 1.000, *g* = 0.80).

## Discussion

The aim of this study was to examine the isometric and dynamic performance adaptations after a 10-week training program that

**Table 5**

**Descriptive strength, sprint, and change of direction data for each training group.\***

| Variable                      |                   | CATCH |      | PULL  |      | OL    |      |
|-------------------------------|-------------------|-------|------|-------|------|-------|------|
|                               |                   | Pre   | Post | Pre   | Post | Pre   | Post |
| IMTP PF (N·kg <sup>-1</sup> ) | Mean              | 36.7  | 35.6 | 38.5  | 42.0 | 37.3  | 42.2 |
|                               | SD                | 5.2   | 5.7  | 5.1   | 6.9  | 11.9  | 13.5 |
|                               | Pre-post <i>g</i> | −0.19 |      | 0.55  |      | 0.37  |      |
| 1RM PC (kg·kg <sup>-1</sup> ) | Mean              | 1.20  | 1.24 | 1.19  | 1.24 | 1.25  | 1.34 |
|                               | SD                | 0.16  | 0.13 | 0.18  | 0.17 | 0.15  | 0.15 |
|                               | Pre-post <i>g</i> | 0.26  |      | 0.27  |      | 0.57  |      |
| 10 m (s)                      | Mean              | 1.88  | 1.87 | 1.89  | 1.86 | 1.92  | 1.83 |
|                               | SD                | 0.08  | 0.08 | 0.13  | 0.09 | 0.12  | 0.11 |
|                               | Pre-post <i>g</i> | −0.12 |      | −0.25 |      | −0.74 |      |
| 20 m (s)                      | Mean              | 3.16  | 3.13 | 3.17  | 3.14 | 3.22  | 3.10 |
|                               | SD                | 0.12  | 0.09 | 0.19  | 0.15 | 0.19  | 0.16 |
|                               | Pre-post <i>g</i> | −0.27 |      | −0.17 |      | −0.65 |      |
| 30 m (s)                      | Mean              | 4.37  | 4.33 | 4.38  | 4.35 | 4.45  | 4.30 |
|                               | SD                | 0.15  | 0.14 | 0.26  | 0.23 | 0.27  | 0.24 |
|                               | Pre-post <i>g</i> | −0.26 |      | −0.12 |      | −0.56 |      |
| 505R (s)                      | Mean              | 2.32  | 2.28 | 2.34  | 2.26 | 2.33  | 2.24 |
|                               | SD                | 0.14  | 0.11 | 0.16  | 0.14 | 0.13  | 0.05 |
|                               | Pre-post <i>g</i> | −0.30 |      | −0.50 |      | −0.87 |      |
| 505L (s)                      | Mean              | 2.29  | 2.28 | 2.34  | 2.26 | 2.37  | 2.25 |
|                               | SD                | 0.08  | 0.12 | 0.19  | 0.13 | 0.11  | 0.09 |
|                               | Pre-post <i>g</i> | −0.09 |      | −0.46 |      | −1.14 |      |

\*CATCH = catch group; PULL = pull group; OL = overload group; Pre = preintervention testing session; Post = postintervention testing session; IMTP = isometric midhigh pull; PF = peak force; 1RM = one repetition maximum; PC = power clean; 505R = right leg 505 change of direction; 505L = left leg 505 change of direction; Pre-Post *g* = preintervention to postintervention Hedge's *g* effect size.

included weightlifting catching or pulling derivatives. An additional goal of this study was to examine the effect of providing a force- and velocity-specific overload stimulus using weightlifting pulling derivatives. In line with our hypotheses, statistically significant differences existed between groups for relative IMTP peak force, 10-, 20-, and 30-m sprint performance, and 505L performance with effect sizes ranging from moderate to large between the OL group and the CATCH and PULL groups. Although no statistical difference in the percent change in 1RM PC or 505R existed between groups, moderate effect sizes were still present, indicating that meaningfully greater effects were produced by the OL group. Also in line with our hypotheses, no statistical or practically meaningful differences existed between the CATCH and PULL groups; the only exceptions were the large and moderate effects that favored the PULL group during the IMTP and 505L tests, respectively.

IMTP peak force is an effective measure of isometric strength (6) that has a moderate to large relationship with a variety of performance characteristics such as sprinting, change of direction, jumping, etc. (46). The OL group in the current study produced the greatest improvements in relative IMTP peak force (13.8%) and displayed large and small practical differences when compared with the CATCH (−2.9%) and PULL (9.0%) groups, respectively. Heavier loading in the midhigh position during certain weightlifting pulling derivatives throughout the training program may have contributed to the improvements of the OL group. For example, the OL group used up to 135, 110, and 102.5% of their PC 1RM during the midhigh pull, counter-movement shrug, and clean pull from the floor, respectively. In addition to the potential for greater positional strength gains, the supramaximal loads used during the OL program likely required greater propulsive forces during the second pull phase of each

**Table 6****Comparison of the volume load (Mean  $\pm$  SD) completed by each training group.\***

| Week  | CATCH                | PULL                     | OL                   | <i>p</i> | CATCH-PULL <i>g</i> | CATCH-OL <i>g</i> | PULL-OL <i>g</i> |
|-------|----------------------|--------------------------|----------------------|----------|---------------------|-------------------|------------------|
| 1     | 21,924 $\pm$ 4,476   | 22,017 $\pm$ 3,830       | 21,889 $\pm$ 3,902   | 0.998    | 0.02                | 0.01              | 0.03             |
| 2     | 23,192 $\pm$ 4,617   | 23,177 $\pm$ 3,410       | 23,439 $\pm$ 3,752   | 0.988    | 0.00                | 0.06              | 0.07             |
| 3     | 25,779 $\pm$ 6,883   | 23,912 $\pm$ 3,593       | 24,822 $\pm$ 3,901   | 0.735    | 0.32                | 0.16              | 0.23             |
| 4     | 16,544 $\pm$ 3,549   | 16,173 $\pm$ 2,698       | 18,361 $\pm$ 3,246   | 0.314    | 0.11                | 0.51†             | 0.70†            |
| 5     | 17,933 $\pm$ 3,607   | 17,285 $\pm$ 2,549       | 19,497 $\pm$ 3,106   | 0.319    | 0.20                | 0.44              | 0.74†            |
| 6     | 18,985 $\pm$ 3,869   | 18,102 $\pm$ 2,778       | 20,611 $\pm$ 3,268   | 0.288    | 0.25                | 0.43              | 0.79†            |
| 7     | 13,759 $\pm$ 2,618   | 13,451 $\pm$ 1,711       | 14,176 $\pm$ 2,505   | 0.802    | 0.13                | 0.15              | 0.32             |
| 8     | 16,652 $\pm$ 3,560   | 15,764 $\pm$ 2,411       | 17,050 $\pm$ 3,342   | 0.678    | 0.28                | 0.11              | 0.42             |
| 9     | 5,899 $\pm$ 1,279    | 5,680 $\pm$ 882          | 5,646 $\pm$ 993      | 0.861    | 0.19                | 0.21              | 0.03             |
| 10    | 4,252 $\pm$ 1,271    | 3,680 $\pm$ 556          | 3,663 $\pm$ 653      | 0.294    | 0.55†               | 0.56†             | 0.03             |
| Total | 164,920 $\pm$ 34,282 | 159,241.1 $\pm$ 22,668.6 | 169,152 $\pm$ 27,414 | 0.331    | 0.19                | 0.13              | 0.38             |

\*CATCH = catch group; PULL = pull group; OL = overload group.

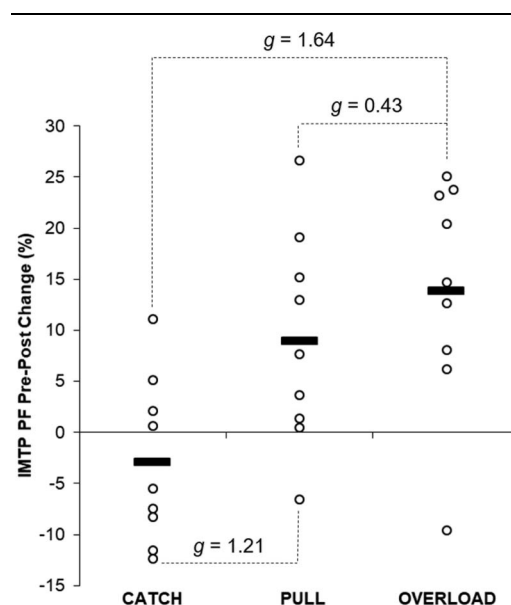
†Moderate effect.

derivative, which may have led to greater force output (8,9). Similar to the OL group, there was a large practical difference between the PULL and CATCH groups. These findings are in contrast to a recent study that compared training with load-matched catching or pulling derivatives 2 days per week in-season for 8 weeks (7). Beyond the potential fatigue effects of in-season training, the differences displayed in the current study may have been due to greater variation in exercise selection and phases of training and the longer duration of the present intervention. It is interesting that the CATCH group, on average, decreased their relative IMTP peak force; however, this may be due to the effort put forth by the subjects during the second pull of their derivatives during their training program. Results from a recent study demonstrated that maximal effort PCs result in greater lower extremity work compared with minimal height PCs (13). Because of the exclusion of the catch phase, the PULL and OL groups may have been able to emphasize the second pull phase of each derivative. It has been reported in previous studies that greater forces are applied in the last 85–100% of the second pull phase during pulling derivatives compared with catching derivatives (27,47). The previous findings suggest that in preparation to catch the barbell, individuals may have less intent to maximize their second pull effort, especially when submaximal loads are used. Collectively, the current results suggest that weightlifting pulling derivatives may provide a greater stimulus for isometric peak force production. Furthermore, a greater benefit may be provided by prescribing loads in excess of a 1RM catching derivative when implementing certain pulling derivatives (e.g., midhigh pull, countermovement shrug, clean pull from floor, etc.).

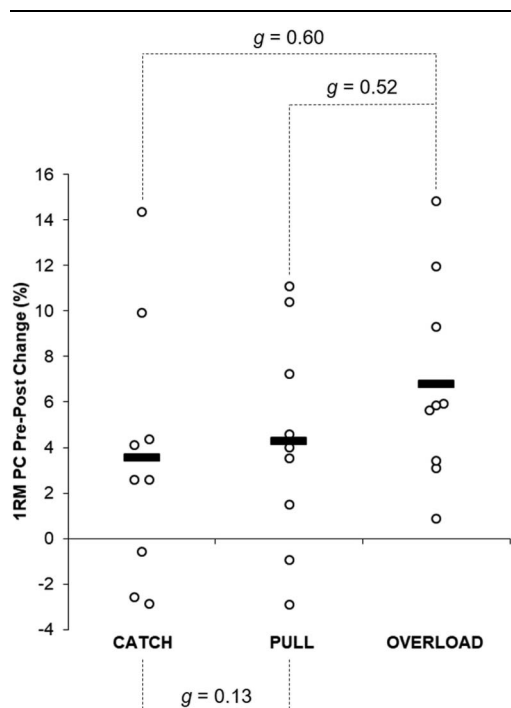
Relative 1RM hang PC strength has been correlated with superior sprint and jump performance (26), which is likely due to similar movement characteristics. The greatest increase in relative 1RM PC performance was produced by the OL group (6.8%), which was followed by the PULL (4.3%) and CATCH (3.5%) groups. Comfort et al. (7) reported no statistical or practically meaningful difference in 1RM PC changes after an 8-week training program that featured load-matched weightlifting catching or pulling derivatives, in line with the comparisons between the CATCH and PULL groups in this study. However, the results of the current study show moderately greater increases in relative 1RM PC in the OL group compared with the other 2 groups. A potential issue that arises with heavier loads during weightlifting catching derivatives is that the athlete may not achieve full hip and knee extension in preparation to drop under

and catch the barbell (27,40,47). Recent literature indicated that maximum effort PCs may increase lower extremity work, knee extensor work, and knee joint excursion compared with a minimal height PC (13). The previous authors noted that maximal effort during the second pull (i.e., triple extension) may also elevate the barbell to a greater extent. Because weightlifting pulling derivatives emphasize the second pull phase, it is possible that the PULL and OL groups may have been able to elevate the barbell to a greater extent during their postintervention testing. Combined with heavier loading, the OL group may have been able to optimize their postintervention 1RM PC adaptations. It should be noted that several of the subjects within the PULL and OL groups mentioned that the PC catch felt “strange,” “awkward,” or “unnatural” during their postintervention 1RM test. However, this may be due to the fact that neither group performed the catch phase nor front squat for the duration of the 10-week program.

The theory behind implementing weightlifting derivatives to improve sprint performance has previously been discussed (14).



**Figure 2.** Percent change in relative isometric midhigh pull peak force (IMTP PF) from preintervention to postintervention with Hedge's *g* comparisons between groups. Bold line denotes group average, and open circles denote individual changes.



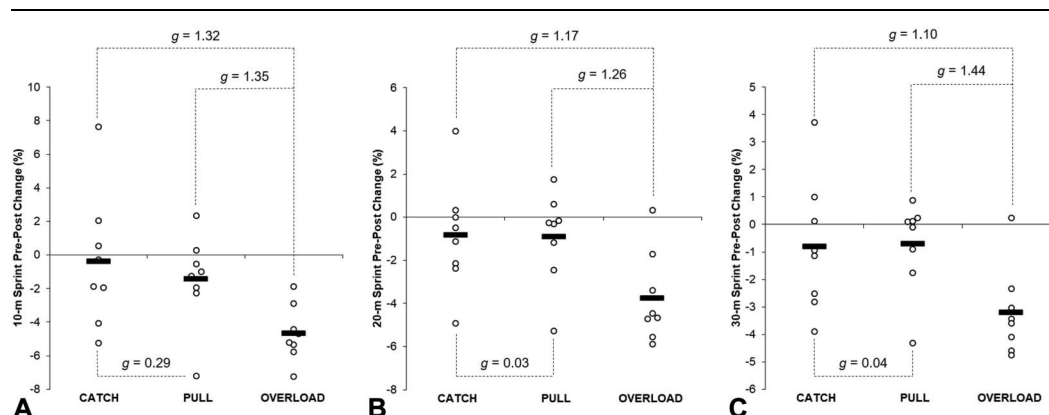
**Figure 3.** Percent change in relative one repetition maximum power clean (1RM PC) from preintervention to postintervention with Hedge's *g* comparisons between groups. Bold line denotes group average, and open circles denote individual changes.

Specifically, weightlifting derivatives may provide a unique training stimulus that may be used enhance both rate of force development and power characteristics. Moreover, these exercises can be programmed in a phase-specific manner to not only enhance the desired fitness characteristics, but also mimic joint angles that are common during various sprint phases. The sprint distances examined within the current study are classified as accelerations given that athletes may require distances longer than 30 m to reach their maximum speed (2). To accelerate effectively, athletes must produce large impulses through a combination of large forces during longer ground contact times (14,24). Although trivial to small effects existed between the CATCH and PULL groups at each sprint distance, the OL group displayed large improvements compared with the other 2 groups. Weightlifting pulling derivatives

may produce greater impulses during the second pull phase compared with catching derivatives (27,47). This is likely due to a greater emphasis on accelerating throughout the second pull phase and omitting the need to drop under and catch the barbell. Thus, an emphasis on the triple extension movement, as well as heavier loading, may have contributed to the improvements in sprint performance by the OL group. Practically speaking, weightlifting derivatives (catching and pulling) may be implemented to help improve accelerative sprint performance. However, it seems that exercises that provide a large force overload stimulus may produce superior training effects. Although the current study focused on accelerative sprint performance, future research should consider examining the effect of weightlifting derivatives on sprint performance over longer distances.

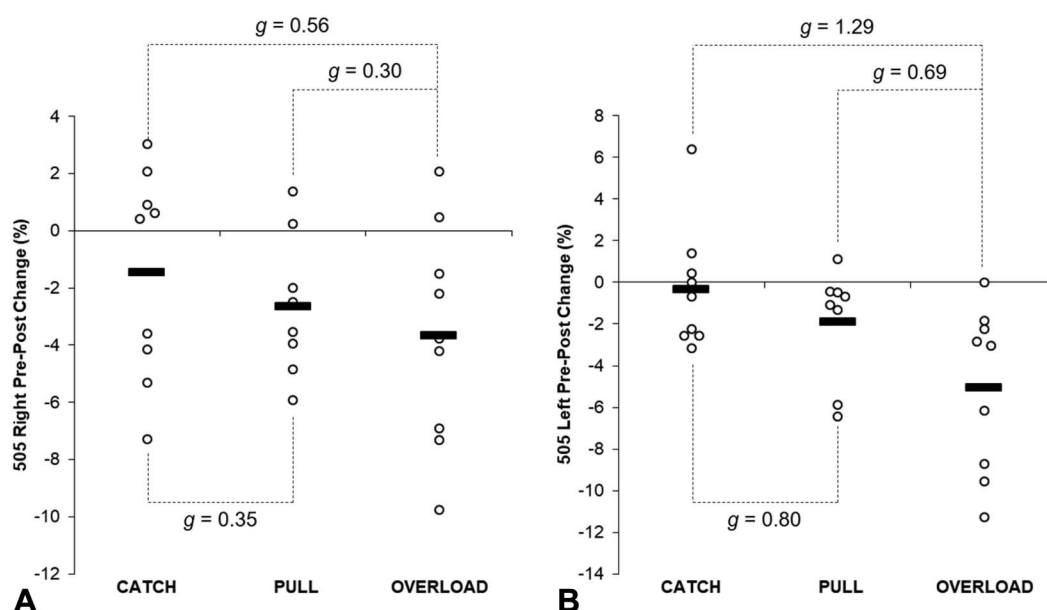
The 505 test has been described as a reliable method that assesses change of direction ability on both legs (1), which is a frequent physical component of many sports (e.g., stop and go movements, cutting, etc.). Similar to the other performance tests, the OL group produced the greatest improvements in both 505R (3.7%) and 505L (5.1%) performance. These results were followed in order by the PULL (505R = 2.6% and 505L = 1.9%) and CATCH groups (505R = 1.5% and 505L = 0.3%). Previous literature indicated that athletes with faster 505 times possess greater eccentric and isometric strength (34) but may also produce greater horizontal propulsive and braking forces (21). As shown above, the OL group produced greater improvements in isometric and dynamic strength, which may have contributed to their 505 improvements. Although not measured in the current study, additional literature indicated that weightlifting pulling derivatives may require similar (10) or greater (10,43,44) work to be performed during the load absorption phase compared with catching derivatives. Thus, it is possible that the use of weightlifting pulling derivatives with heavier loads during the OL program may have contributed to a greater capacity to absorb force and create larger braking forces during the 505 test. As mentioned above, larger propulsive impulses during pulling derivatives may have also contributed to the current results. Despite the current findings, it should be noted that additional literature has suggested that motor control and coordination may be the primary factors that contribute to 505 performance (52). Thus, further analysis of change of direction characteristics after training programs that implement weightlifting derivatives may be warranted.

When implementing weightlifting movements into resistance training programs, it is important to prescribe an



**Figure 4.** Percent change in 10- (A), 20- (B), and 30-m (C) sprint performance from preintervention to postintervention with Hedge's *g* comparisons between groups. Bold line denotes group average, and open circles denote individual changes.





**Figure 5.** Percent change in 505 change of direction performance on the right (A) and left (B) legs from preintervention to postintervention with Hedge's  $g$  comparisons between groups. Bold line denotes group average, and open circles denote individual changes.

exercise and load combination that will match the fitness demands of each training phase. Interestingly, no statistically significant differences existed between groups when comparing the volume-load completed. It should be noted, however, that moderate effect sizes were present when comparing the volume-load completed by the OL group and CATCH and PULL groups during the max-strength and speed-strength phases of the study. A primary benefit of prescribing weightlifting pulling derivatives is that the exercises allow for a wider spectrum of loads to be prescribed. Although catching derivatives are limited to their 1RM on the high load end of the spectrum, loads for pulling derivatives may exceed the 1RM PC as discussed above or increase up to 140% 1RM as shown in previous literature (8,9,30). On the low load end of the spectrum, it is difficult for athletes to maximize their effort when they perform the second pull during catching derivatives due to the potential to “overpull” the barbell, which may lead to poor technique during the catch phase. The lowest loads used for pulling derivatives in the current study were 30 and 35% 1RM for the jump shrug and hang high pull, respectively, which was in line with previous literature for peak power development (28,35,37). Because maximal effort can be given on both ends of the loading spectrum while providing a force and velocity overload stimulus, it seems that implementing pulling derivatives may be highly beneficial to resistance training programs. It should be noted that the findings of the current study do not discount the effectiveness of training with weightlifting catching derivatives because a number of studies have shown how beneficial they are compared with other training methods (31,50,51). Although the current study compared only catching or pulling derivatives within a training program, it is possible and encouraged to implement both variations when training athletes. In fact, weightlifting catching derivatives may provide a similar training stimulus to load-matched pulling derivatives (4,5,7). Thus, both types of

derivatives may be used interchangeably based on the goals of each fitness phase. For an example of how implement both weightlifting catching and pulling derivatives in the same program, readers are directed to a previous review (39).

A potential limitation to the current study is the fact that each weightlifting derivative was programmed based on each subjects' preintervention 1RM PC. If the PC is regularly prescribed in training, the use of this method may not be detrimental. However, if an individual does not perform a 1RM PC, practitioners may find it difficult to prescribe loads for pulling derivatives. Only one study has examined an alternative method of loading for a weightlifting pulling derivative (e.g., percentage of body mass) (45), and thus, further research on this topic is warranted. A second limitation may have been the length of the overall training program. Although the 10-week program allowed for strength-endurance, strength, overreach, and taper phases to take place, low-repetition strength work (e.g., 3 sets of 3 repetitions at 85% 1RM or higher) was not performed. This was in part due to the length of the academic semester and the need to work around breaks during the academic year. Although each subject experienced the same volume within each training block, the CATCH and PULL group did not experience loads greater than 82.5% of their 1RM PC during their prescribed weightlifting exercises. Although this may have contributed to the lack of improvement in relative IMTP peak force for the CATCH group, it should be noted that both the weakest and strongest individuals (based on relative squat strength) within the group decreased their relative IMTP peak force by at least 7.5%. Furthermore, while 5 of the 9 subjects in the CATCH group decreased their relative IMTP peak force, only one individual in the PULL group failed to improve their performance. Finally, the volume-load completed by each group may be listed as a limitation (albeit a necessary one). A purpose of this study was to examine the effect of manipulating exercises and load using weightlifting

derivatives to benefit strength-power characteristics. The current results indicate that a benefit of implementing weightlifting pulling derivatives is the ability to prescribe loads that emphasize either force (heavier loads) or velocity (lighter loads), which may modify the overall volume-load completed. From a training-efficiency standpoint, it is recommended that future research should continue to examine the relationship between performance changes and volume-load when using weightlifting derivatives. Specifically, researchers should consider examining volume-load calculated using the displacement of the barbell. Although not examined in the current study, it may be argued that weightlifting pulling derivatives performed at the same loads (or heavier) as catching derivatives may produce a lower overall volume-load given that the barbell displacement for certain exercises (e.g., midhigh pull, pull from the floor, etc.) is smaller and, thus, may be more efficient at producing a strength-power stimulus compared with catching derivatives.

### Practical Applications

The findings of the current study indicate that weightlifting catching and pulling derivatives may improve a variety of isometric and dynamic performance characteristics. However, it seems that training with a force- and velocity-specific overload stimulus using weightlifting pulling derivatives may produce superior training effects compared with submaximal load-matched weightlifting catching and pulling derivatives. It should be noted that submaximally loaded pulling derivatives may also produce superior performance gains compared with using catching derivatives at the same relative loads when it comes to relative IMTP peak force. Practitioners should consider implementing weightlifting pulling derivatives to expand the loading spectrum that an athlete can experience within their training program. Specifically, it may be beneficial from a force production standpoint to implement loads in excess of an athlete's 1RM PC, but also lighter, submaximal loads to provide a greater velocity stimulus. However, it is important to match the demands of each fitness phase by prescribing the most effective exercise and load combinations. Although weightlifting pulling derivatives may have the potential to maximize adaptations on the heavy- and light-load ends of the loading spectrum, it is important to note that weightlifting catching derivatives may be effectively implemented with pulling derivatives rather than prescribing only one method or the other.

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